

VIA HOLE TYPES And Their USE IN PCBs

The importance of via holes cannot be ignored by the designers of multi-layered PCBs. This article describes various types of via holes and how they help in proper designing of the PCBs



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It is a well-known fact that a PCB is the heart of an electronic system. Hardware engineers have lots of expectations from PCB designers with respect to signal integrity, power integrity, electromagnetic compatibility (EMC) as well as thermal analysis. Depending upon the type of circuit, all these parameters need to be managed in a PCB design so as to make an assembly successful for its intended use.

In this digital world the signal speed ranges from a few MHz to several GHz. With the increased demand of functions per assembly/chip and miniaturisation of electronic systems, PCBs are becoming more and more complex. The increasing requirement of multiple functions in a chip is resulting in complex surface mounted device (SMD) components, which have multiple pins placed close together or are even leadless packages. The multi-pin components like ball grid arrays (BGAs) and thin quad flat packages

(TQFPs) force the PCB designers to go for multilayer PCBs.

Via holes play a major role while designing the multilayer PCBs. A via hole is a plated through hole (PTH), which is used to transfer a signal from one layer of a PCB to another layer, without the scope of mounting any component into these holes.

Most engineers make use of via holes just to transfer signals. But there are other features too that are associated with these via holes. These via holes can be used in various ways. A via hole can have a positive impact when it comes to route a signal (interconnection) and a negative impact when it comes to route a sensitive signal.

Various type of via holes

Through hole via, where one can see through a PCB. Such a via hole starts from the top layer of a PCB and ends at the bottom layer. It must be defined in each and every electrical layer of a PCB.

Blind via, whose starting point is visible, but end point is not visible. Such via holes are used in multilayer PCBs. A blind via starts from an outer layer and disappears in inner layers.

Buried via, which is not visible at all. It is used to transfer a signal from one inner layer to another inner layer. A buried via is used to save the space for signal routing over outer or inner layers.

Micro via, which has a comparatively much smaller drill diameter. Due to shrinkage of electronic assemblies and routing of high-density-intercon-

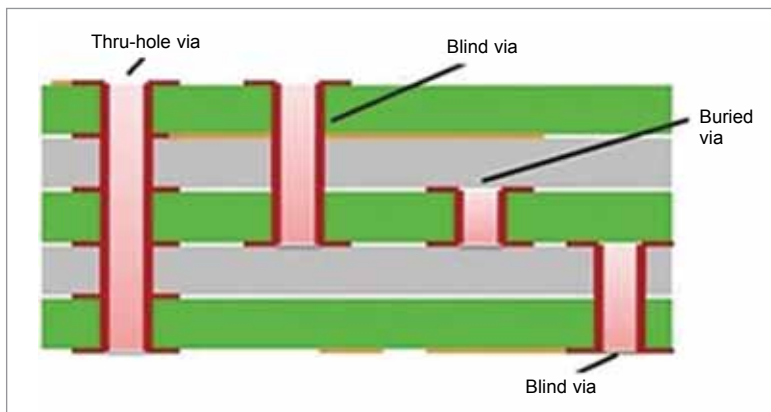


Fig. 1: Illustration of some via hole types